UTF-8 and D3.js

D3 utilizes UTF-8 characters in its code, which means that you can do one of three things to make sure you don't have any errors.

You can set your document to UTF-8:

```
<!DOCTYPE html><meta charset="utf-8">
```

Or you can set the charset of the script to UTF-8:

```
<script charset="utf-8" src="d3.js"></script>
```

Or you can use the minified script, which shouldn't have any UTF-8 characters in it:

```
<script src="d3.min.js"></script>
```

Three categories of information about each element determine its behavior and appearance: styles, attributes, and properties. *Styles* can determine transparency, color, size, borders, and so on. *Attributes* typically refer to classes, IDs, and interactive behavior, though some attributes can also determine appearance, depending on which type of element you're dealing with. *Properties* typically refer to states, such as the "checked" property of a check box, which is `true` if the box is checked and `false` if the box is unchecked. D3 has three corresponding functions to modify these values. If we wanted to modify the HTML elements in the previous example, we could use D3 functions that abstract this process:

```
d3.select("#someDiv").style("border", "5px darkgray dashed");
d3.select("#someDiv").attr("id", "newID");
d3.select("#someCheckbox").property("checked", true);
```

Like many D3 functions of this kind, if you don't signify a new value, then the function returns the existing value. You'll see this in action throughout this book, and later in the chapter as you write more code, but for now remember that these three functions allow you to change how an element appears and interacts.

The DOM also determines the onscreen drawing order of elements, with child elements drawn after and inside parent elements. Although you have some control over drawing elements above or below each other with traditional HTML using `z-index`, this isn't available for SVG elements (though it might be implemented at some point using the `render-order` attribute).

EXAMINING THE DOM IN THE CONSOLE

Navigate to d3ia.html, and you can get exposure to how D3 works. The page isn't very impressive, with just a single, black-outlined rectangle. You could modify the look and feel of this web page by updating d3ia.html, but you'll find that it's easy to modify the page by using your web browser's developer console. This is useful for testing changes to classes or elements before implementing them in your code. Open up the developer console, and you'll have two useful screens, shown in figures 1.11 and 1.12, which we'll go back to again and again.